

# Zhiyuan Yao

Postdoctoral Researcher, Hebrew University of Jerusalem  
Curriculum Vitae

## Contact Information

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## Education

**Postdoc, Hebrew University of Jerusalem, Israel**

10/2022 - Present  
Advisor: Nir Mandelker, Avishai Dekel

**Joint Ph.D, Max Planck Institute for Astrophysics, Germany**

04/2021 - 03/2022  
Advisor: Volker Springel, Max Gronke

**Ph.D, Astronomy, Shanghai Astronomical Observatory, Chinese Academy of Sciences, China**

09/2016 - 06/2022  
Advisor: Feng Yuan

**Bachelor, Astronomy, University of Science and Technology of China, China**

09/2012 - 06/2016

## Research Interests

Galaxy formation and evolution; circumgalactic medium; multiphase hydrodynamics; cold streams and virial shocks; turbulence and mixing; numerical simulations; high-redshift galaxies; computational astrophysics

## Honors & Awards

2024, Rosenblum prize  
2021-2022, Sino-German Joint PhD Exchange Program (Shanghai Astronomical Observatory and Max Planck Institute for Astrophysics, 04/2021 - 03/2022)  
2020, Academic Scholarship of University of Chinese Academy of Sciences

## Selected Publications

### First-author publications

**Yao Z.**, Mandelker N., Oh S. P.

“Cold stream penetration of virial shocks: fragmentation, coagulation, and disruption in the hot circumgalactic medium”, submitted to MNRAS, arXiv:2607.14090 (2026)

**Yao Z.**, Mandelker N., Oh S. P., Aung H., Dekel A.

“Effects of cloud geometry and metallicity on shattering and coagulation of cold gas, and implications for cold streams penetrating virial shocks”, MNRAS, 536, 3053 (2025)

**Yao Z.**, Yuan F., Ostriker J. P.

“Active galactic nucleus feedback in an elliptical galaxy with updated AGN physics: parameter exploration”, MNRAS, 501, 398 (2021)

**Yao Z. & Gan Z.**

“Hot gas flows on a parsec scale in the low-luminosity active galactic nucleus NGC 3115”, MNRAS, 492, 444 (2020)

## Collaborative publications

Chen, H.-Z., Li, Z., Dekel, A., **Yao Z.** et al.

“Feedback-Free Star Formation in Clusters within a Galaxy Simulated at High Resolution in Cosmic Dawn”, arXiv:2606.12605 (2026)

Dekel, A., Dutta Chowdhury, D., Lapiner, S., **Yao Z.** et al.

“From Feedback-Free Star Clusters to Little Red Dots via Compaction”, arXiv:2511.07578 (2025)

Dekel, A., Mandelker, N., Li, Z., **Yao Z.** et al.

“From FFB Starbursts at Cosmic Dawn to Quenching at Cosmic Morning: Hi-z Galaxy Bimodality”, MNRAS, 544, 160 (2025)

Yang, G. et al. (incl. **Yao Z.**)

“Do current X-ray observations capture most of the black-hole accretion at high redshifts?”, ApJ, 921, 170 (2021)

## Invited & Seminar Talks

12/2025, Tel Aviv University Seminar, Tel Aviv, Israel

10/2021, MPA Institute Seminar, Garching, Germany

## Conference Contributions

07/2026, The 71th Annual Meeting of the Israel Physics Society, Jerusalem, Israel

11/2025, Modelling of Multiphase Astrophysical Media, Ringberg, Germany

08/2025, Galaxy Formation Workshop, Santa Cruz, United States

07/2025, Cosmic Ecosystems, Waterloo, Canada

08/2024, Galaxy Formation Workshop, Santa Cruz, United States

05/2023, Modelling of Multiphase Astrophysical Media, Kochel, Germany

08/2022, Galaxy Formation Workshop, Santa Cruz, United States

07/2021, Ringberg Retreat, Munich, Germany

Earlier (2018–2020): 4 contributed talks at meetings in China and Germany.

## Technical Skills

Hydrodynamic simulation codes: AREPO, RAMSES, ZEUS-MP/2

Programming: Python, C, C++, Fortran, MPI/OpenMP

Analysis tools: yt, NumPy, SciPy, Matplotlib

High-performance computing: distributed-memory simulations on large HPC systems

## Mentoring

2025 – Aviad Leibovich (B.Sc. student), Hebrew University of Jerusalem

## Professional Service

Organizer, Astrolunch Seminar Series, Hebrew University of Jerusalem

## References

Nir Mandelker — Hebrew University of Jerusalem — nir.mandelker@mail.huji.ac.il

Feng Yuan — Fudan University — fyuan@fudan.edu.cn

Volker Springel — Max Planck Institute for Astrophysics — vspringel@mpa-garching.mpg.de